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Dato: 15/04/2026

Trigonometri II

Opgave 4.10a

$A = 70^\circ$
$a = 7\ cm$
$b = 5\ cm$

Man kan regne B ved hjælp af sinusrelationen:

$\frac{\sin(B)}{b} = \frac{\sin(A)}{a}$
$\Updownarrow$
$B = \arcsin\left(\frac{\sin(A)}{a} \cdot b\right) = \underline{\underline{42,16^\circ}}$

C beregnes ved hjælp af vinkelsummen i trekanter:

$C = 180^\circ - A - B = \underline{\underline{67,84^\circ}}$

Man kan beregne c ved hjælp af sinusrelationen:

$c = \frac{a}{\sin(A)} \cdot \sin(C) = \underline{\underline{6,899\ cm}}$

Opgave 4.10b

$A = 41,6^\circ$
$B = 42,3^\circ$
$a = 12,3\ cm$
$C = 180^\circ - A - B = \underline{\underline{96,1^\circ}}$
$c = \frac{a}{\sin(A)} \cdot \sin(C) = \underline{\underline{18,42\ cm}}$
$b = \frac{a}{\sin(A)} \cdot \sin(B) = \underline{\underline{12,47\ cm}}$

Opgave 4.10c

$C = 20,3^\circ$

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$b = 5\ cm$
$c = 2,5\ cm$
$B = \arcsin\left(\frac{\sin(C)}{c} \cdot b\right) = \underline{\underline{43,94^\circ}}$
$A = 180^\circ - C - B = \underline{\underline{115,8^\circ}}$
$a = \frac{b}{\sin(B)} \cdot \sin(A) = \underline{\underline{6,49\ cm}}$

Opgave 4.13a

$a = 10\ cm$
$b = 8\ cm$
$C = 60^\circ$

Her bruges cosinusrelationen til at beregne c:

$c^2 = a^2 + b^2 - 2 \cdot a \cdot b \cdot \cos(C)$
$\Updownarrow$
$c = \sqrt{a^2 + b^2 - 2 \cdot a \cdot b \cdot \cos(C)} = \underline{\underline{9,165\ cm}}$

Derefter behøves man ikke bruge cosinusrelationen, da vi nu kan bruge sinusrelationen.

$B = \arcsin\left(\frac{\sin(C)}{c} \cdot b\right) = \underline{\underline{49,11^\circ}}$
$A = 180^\circ - B - C = \underline{\underline{70,89^\circ}}$

Opgave 4.13b

$A = 120^\circ$
$b = 4,6\ cm$
$c = 5,8\ cm$

Samme koncept som i 4.13a, her bruges cosinusrelationen til at finde a.

$a^2 = b^2 + c^2 - 2 \cdot b \cdot c \cdot \cos(A)$
$\Updownarrow$
$a = \sqrt{b^2 + c^2 - 2 \cdot b \cdot c \cdot \cos(A)} = \underline{\underline{9,027\ cm}}$

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Sinusrelationen for at finde B:

$B = \arcsin\left(\frac{\sin(A)}{a} \cdot b\right) = \underline{\underline{26,19^\circ}}$
$C = 180^\circ - A - B = \underline{\underline{33,81^\circ}}$

Opgave 4.13c

$a = 13,6\ mm$
$b = 15,9\ mm$
$c = 16,7\ mm$

Her findes A ved brug af cosinusrelationen:

$A = \arccos\left(\frac{b^2+c^2-a^2}{2 \cdot b \cdot c}\right) = \underline{\underline{49,24^\circ}}$

Nu kendes nok til at bruge sinusrelationen:

$B = \arcsin\left(\frac{\sin(A)}{a} \cdot b\right) = \underline{\underline{62,32^\circ}}$
$C = \arcsin\left(\frac{\sin(A)}{a} \cdot c\right) = \underline{\underline{68,45^\circ}}$

Man kan verificere at svaret er korrekt, da alle vinklerne lagt sammen giver 180 grader.

$A + B + C = 180^\circ$

Opgave 4.15a

$a = 6,82\ m$
$b = 12,46\ m$
$B = 123,3^\circ$
$A = \arcsin\left(\frac{\sin(B)}{b} \cdot a\right) = 27,22^\circ$
$C = 180^\circ - A - B = 29,48^\circ$
$c = \frac{b}{\sin(B)} \cdot \sin(C) = 7,335\ m$

Man kan bruge Herons formel for at finde arealet. Der startes med at beregn trekants halve omkreds (s):

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$s = \frac{1}{2} \cdot (a + b + c) = 13,31\ m$

Derefter bruges Herons formel:

$A_1 = \sqrt{s \cdot (s-a) \cdot (s-b) \cdot (s-c)} = \underline{\underline{20,91\ m^2}}$

Opgave 4.15b

Variabler:

$a = 8\ cm$
$b = 11\ cm$

$c = 13\ cm$
$A = \arccos\left(\frac{b^2+c^2-a^2}{2 \cdot b \cdot c}\right) = 37,79^\circ$
$B = \arcsin\left(\frac{\sin(A)}{a} \cdot b\right) = 57,42^\circ$
$C = \arcsin\left(\frac{\sin(A)}{a} \cdot c\right) = 84,78^\circ$
$s = \frac{1}{2} \cdot (a + b + c) = 16\ cm$
$A_1 = \sqrt{s \cdot (s-a) \cdot (s-b) \cdot (s-c)} = \underline{\underline{43,82\ cm^2}}$

Opgave 4.15c

$A = 127^\circ$
$b = 4\ cm$
$c = 6\ cm$
$a = \sqrt{b^2 + c^2 - 2 \cdot b \cdot c \cdot \cos(A)} = 8,994\ cm$
$B = \arcsin\left(\frac{\sin(A)}{a} \cdot b\right) = 20,81^\circ$
$C = \arcsin\left(\frac{\sin(A)}{a} \cdot c\right) = 32,19^\circ$
$s = \frac{1}{2} \cdot (a + b + c) = 9,497\ cm$
$A_1 = \sqrt{s \cdot (s-a) \cdot (s-b) \cdot (s-c)} = \underline{\underline{9,584\ cm^2}}$

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Opgave 4.22

$k = 80\ cm$
$h = 20\ cm$

Man kan bruge Pythagoras læresætning for at beregne hvad r er. Forestil radius er grundlinjen, eller den ene katete, radius minus højden er den anden katete.

$r^2 = \left(\frac{k}{2}\right)^2 + (r-h)^2$
$\Updownarrow$

Udvider ved brug af 2. kvadratsætning:

$r^2 = \left(\frac{k}{2}\right)^2 + r^2 - 2 \cdot r \cdot h + h^2$
$\Updownarrow$

Trække r i anden fra begge sider:

$0 = \left(\frac{k}{2}\right)^2 - 2 \cdot r \cdot h + h^2$
$\Updownarrow$

Isolere r:

$2 \cdot r \cdot h = \left(\frac{k}{2}\right)^2 + h^2$
$\Updownarrow$
$2 \cdot r = \left(\frac{k}{2}\right)^2 + h$

2r er lig med diameteren:

$d = \frac{k^2}{4h} + h = \underline{\underline{100\ cm}}$

Opgave 4.24

Ved brug af pythagoras læresætning kan man finde både a, b og c:

$a = \sqrt{0,9\ m^2 + 1,8\ m^2} = \underline{\underline{1,643\ m}}$
$b = \sqrt{(3,1\ m - 0,9\ m)^2 + 1,8\ m^2} = \underline{\underline{2,577\ m}}$
$c = \sqrt{2,4\ m^2 + 3,1\ m^2} = \underline{\underline{2,345\ m}}$

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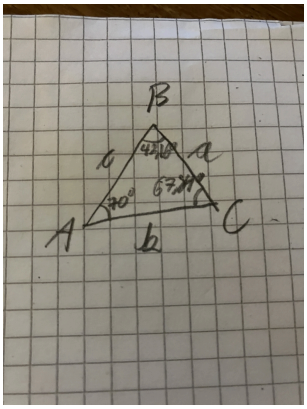
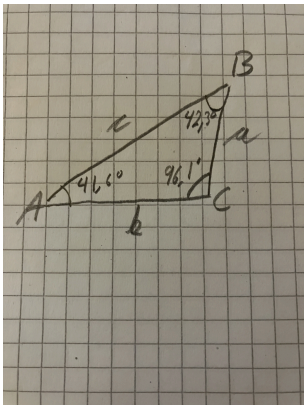
Bilag

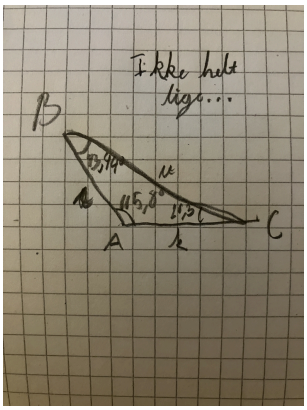
Opgave 4.10a



Opgave 4.10b



Opgave 4.10c



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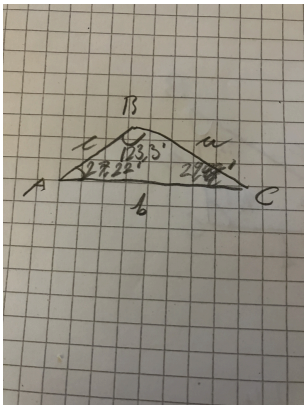
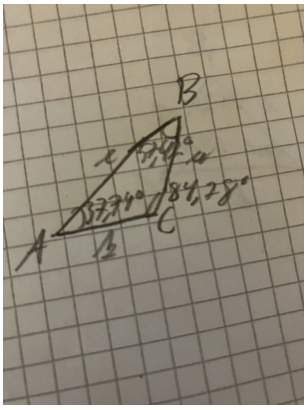
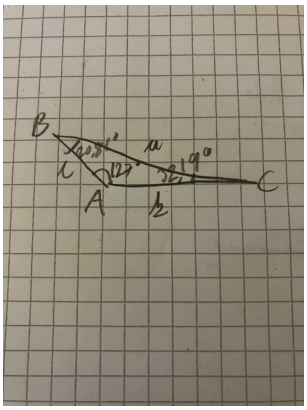
Opgave 4.15a



Opgave 4.15b



Opgave 4.15c